
DAPHNE DAQ Integration

Ethernet data taking, standalone calibration, and pds-run scans

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DUANE
DEEP UNDERGROUND
NEUTRINO EXPERIMENT

 **Fermilab**

Executive Claim

WHAT IS NOW TRUE

DAPHNE Ethernet data taking is established in DAQ; calibration scans are being moved from standalone tools into `pds-run`.

Decoding belongs in DAQ libraries, with Waffles downstream. Calibration policy belongs in `pds-run`. Firmware/schema work adds peak descriptors and the x-correlation self trigger; firmware dead-time remains active.

Nov 2025

DAQ baseline proven in the VD Coldbox.

Feb 2026

Standalone calibration proven in the same VD Coldbox facility.

Mar-Apr 2026

Peak descriptors and x-correlation self trigger implemented; dead-time work ongoing.

May 2026

DAQ-wrapped calibration exercised with `pds-run`.

Scope And Success Criteria

DAQ baseline

Proven in the VD Coldbox, November 2025; decodes DAPHNE Ethernet fragments and preserves monitoring in fddaq-v5.5.0.

Standalone calibration

Proven in the same VD Coldbox facility, February 2026, with direct DAPHNE control and spy-buffer capture.

DAQ-wrapped calibration

May 2026 work: selected scan variables move into `pds-run`; every point creates a DAQ run.

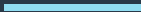
Constraint: the DAQ environment, OKS database, proxy state, and tmux session are part of the apparatus. The change must survive a short run on `np04-srv-024` through `np02-vdcb`.



SECTION

DAQ Baseline

The three DAQ packages define the format, the decode API, and the live controller path



Baseline Data Path



Format

C++ classes describe the wire layout and enforce bit-field boundaries.

Decode

Python bindings expose arrays for analysis without rewriting the unpacker.

Waffles

Waffles consumes the decoded arrays for calibration and validation workflows.

Changed DAQ Repositories

fddetdataformats

`marroyav/daphneeth`

Frame definitions and tests

New: peak descriptor support

Adds DAPHNE ETH stream frame bindings.

rawdatautils

`marroyav/daphneeth`

NumPy-facing unpackers

Exposes ADC, timestamp, channel, stream, and peak descriptor arrays for Waffles and analysis.

daphnmodules

`patch/fddaq-v5.5.x`

DAPHNE V3 controller

Adds V3 GeneralInfo monitoring used during operations.

Data-Format Contract

DAPHNEEthFrame

Single-channel frame: Ethernet header, DAPHNE metadata, five peak descriptors, and 1024 packed 14-bit ADC samples.

Peak descriptors

New firmware information is now persisted in the frame and exposed through DAQ formats and Python decode utilities.

DAPHNEEthStreamFrame

Four-channel stream frame with 280 samples per channel packed as 14-bit ADCs.

Reviewer handles

- ▶ Constants: ADC width, sample count, channel count.
- ▶ Accessors validate channel, sample, and peak indices.
- ▶ Unit tests round-trip ADC and peak fields.
- ▶ Bindings register triggered and stream types.

Decode Contract

RAW DATA UTILS

`rawdatautils` is not a second format definition. It reuses `fdetdataformats` and exposes arrays consumed by Waffles and validation scripts:

- ▶ ADC, timestamp, and channel arrays
- ▶ stream-frame equivalents
- ▶ peak descriptor arrays for triggered frames



Controller And Monitoring Contract

daphnmodules V3 controller

Configures the DAPHNE V3 analog chain, reads counters, and publishes opmon in DAQ.

Important release finding

fdDAQ-v5.5.0-a9-1 ships 3.0.1. V3 GeneralInfo opmon appears in 3.0.2 on patch/fdDAQ-v5.5.x.

ReadTriggerCounters



per-channel
threshold, record, busy, full



ReadGeneralInfo
bias, power, temp



Kafka opmon
shifter view



SECTION

Standalone Calibration

`daphne-server` established fast board-level scans outside DAQ run control



Why Standalone Was Useful



Scientific role

Find settings before DAQ run-control time.

Engineering role

Exercise routes, spy buffers, LED clients, and analysis.

Boundary

Diagnostic data is not a substitute for DAQ-integrated data taking.

Standalone Products

Dark scans

Scan `vgain`/offset and find pedestal regions from spy-buffer captures.

LED scans

Scan LED intensity, mask, and channel sets with direct SSP control.

Bundles

Generate per-`vgain` JSON settings for later replay.

Example dynamic-range result: `vgain` 1500–2500 is useful for the tested LED response; 2750 and 3000 did not reach the 2000 ADC pedestal target in the scanned offset range.

Standalone Command Shape

```
./utils/run_client_via_tunnel.sh daphne-13 \  
  client/configure_fe_from_output.py \  
  client/dynamic_range_led/per_channel_target2k_vgain2000.json  
  
./utils/run_client_via_tunnel.sh daphne-13 \  
  client/protobuf_acquire_list_channels.py --osc_mode \  
  -foldername client/dynamic_range_led/runs/vgain2000_led_sat \  
  -channel_list 4,5,6,7,12,13,14,15,18,19,21 -N 64 -L 2048
```

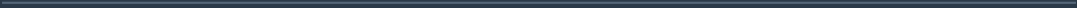
Operational caution: the VD Coldbox target must be checked in configuration. Inventory maps daphne-14 to 10.73.137.160; some historical examples use daphne-13/10.73.137.161.



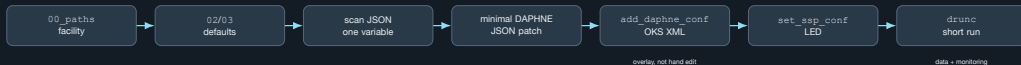
SECTION

DAQ-Wrapped Calibration

`pds-run` moves selected scan variables into the DAQ configuration and run-control chain



What `pds-run` Adds



Model

Paths, commands, defaults, and scan dimensions live in numbered JSON files.

Overlay

DAPHNE changes become minimal JSON patches, then OKS XML.

Execution

Each scan point can set SSP LED state and take a DAQ run.

Configuration Chain

Facility

- ▶ `00_paths.json`: workarea, DB, OKS, target.
- ▶ `01_commands.json`: proxy and helper commands.
- ▶ `02_run_defaults.json`: rate, wait, SSP LED.
- ▶ `03_default_config.json`: base DAPHNE detail.

Scan

- ▶ `thr-scan`, `selfthr-scan`, `att-scan`.
- ▶ `offset-scan`, `trim-scan`.
- ▶ `led-intensity-scan`: masks and intensities.
- ▶ `plan_only`: inspect XML and commands.

Review rule: one scan file explains the changed variable; it does not re-declare the whole DAQ environment.

DAQ Run Command Generated By `pds-run`

```
drunc-unified-shell ssh-CERN-kafka.json \  
  /nfs/sw/dunedaq/dunedaq-fddaq-v5.5.0-dev-pds/.../pds-vdcb-session.data.xml \  
  np02-session pds-vdcb-pds \  
  start-run --run-type TEST \  
  change-rate --trigger-rate 20.0 \  
  wait 120 shutdown terminate
```

Before run

Source DAQ env, source DB path,
disable proxy for runtime.

During run

Run under tmux on the
run-control host.

After run

Confirm data files and opmon.

Operations Commands For pds-run Scans

```
# Plan first: set "plan_only": true in the scan JSON.
pds-run led-intensity-scan configs/vd_coldbox/06_led_calib.json
pds-run led-intensity-scan configs/vd_coldbox/02_run_defaults.json \
  --mask-values 1,2,4,8 \
  --led-intensity-values 3000,3500,4095
pds-run thr-scan configs/vd_coldbox/04_stthre.json
pds-run selfthr-scan configs/vd_coldbox/04_stthre.json
pds-run offset-scan configs/vd_coldbox/<offset_scan>.json
pds-run trim-scan configs/vd_coldbox/<trim_scan>.json
```

Each command updates only the scan variable, applies the minimal DAPHNE XML overlay, runs SSP setup when requested, then delegates to drunc.

Operational Environment

Login path

```
marroyav@np04-srv-017 then  
np04daq@np04-srv-024; use tmux session  
np02-vdcb.
```

Workarea

```
/nfs/sw/dunedaq/  
dunedaq-fddaq-v5.5.0-dev-pds.  
Source setup_for_run.sh.
```

Proxy state

Use `source ~/bin/web_proxy.sh` only for package downloads. Disable for running with `-u`.

Hardware target

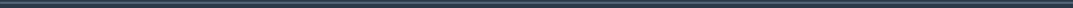
Board identity and IP must be checked in the active JSON/XML; target DAPHNE selection is a configuration requirement, not an assumption.



SECTION

Validation And Review

The success condition is both software cleanliness and live DAQ behavior



Live Validation Result

Run 44355

Default release with no local `daphnmodules`: data taking worked, but V3 monitoring was not visible.

Run 44356

Clean `daphnmodules` 3.0.2 override: two-minute TEST run succeeded at 20 Hz.

Firmware

Peak descriptors were added in firmware and then carried into the DAQ data-format contract.

Conclusion

Monitoring loss was traced to the release package version, not to run control.

Final tested state: `sourcecode/daphnmodules` commit `13289b5` on patch/`fddaq-v5.5.x`; `DAPHNE_V3_ROUTE` unset; monitoring visible.

Live pds-run Scan Exercise

Mode	Runs	Variable exercised
self-trigger	44357-44358	DAPHNE threshold: 11, 10
AFE attenuation	44359-44360	all AFE attenuators: 2200, 2300
offset	44361-44362	selected channel offsets: 2260, 2275
trim	44363-44364	selected channel trims: 1, 0
AFE bias	44365-44366	AFE0 bias: 10, 0; others fixed at 0
LED / SSP	44367-44368	LED intensity: 3000, 4095; mask 4

Branch/host: marroyav/vdcb-daq-scans at 06efc5c on np04-srv-024, tmux np02-vdcb; logs

/tmp/pds_vdcb_live_20260508_152420.

Result: every successful scan command exited with status 0; xmllint -noout passed after each XML update; no successful-run traceback or drunc command failure was found.

Live Exercise Findings

Finding	Reviewer-facing consequence
SSP object name	Active XML uses <code>np02-ssp</code> ; scan configs must not silently rely on the old <code>np02-ssp-on</code> default.
Execution seed	<code>99_apply_fe_min_v2_subset.json</code> is plan-safe but not execution-safe; <code>add_daphne_conf</code> requires <code>detector_id</code> .
Zero/default XML	<code>trim=0</code> , <code>v_bias=0</code> , and <code>bias_ctrl=0</code> can be omitted by <code>add_daphne_conf</code> ; absence of these attributes is not by itself a run failure.
Post-scan state	The exercise left SSP at <code>channel_mask=4</code> , <code>pulse1_width=3</code> ; scan plans should state whether to restore the baseline.

Conclusion: the DAQ-wrapped scan loop is operational for the tested modes; remaining cleanup is configuration hygiene, explicit target naming, and post-scan restoration policy.

Summary

DAQ baseline: proven in the VD Coldbox, November 2025.

Standalone calibration: proven in the same facility, February 2026.

`pds-run`: DAQ-wrapped calibration, May 2026.

Peak descriptors: firmware, DAQ format, decode, Waffles.

Backup: Interface Reference

FORMAL INTERFACE HANDLE

DUNE Interface Control Document: DAQ & Photon Detector System (PDS)

EDMS 2088726 v7

<https://edms.cern.ch/document/2088726/7>

Use in discussion

The ICD is the external reference for the DAQ/PDS boundary. Resource and dead-time work remains an implementation risk, but the mitigation path is constrained by this interface contract.

Backup source for resource evidence: `repo/Latex/daphne_hardware_architecture_may2026`, sections “Baseline PL Resource Report” and “Current Grouped Candidate Build Result”.

Backup: K26 Resource Mitigation Evidence

BASELINE, OUTPUT-2C337B5



GROUPED CANDIDATE, E5E2D70



Interpretation: the grouped candidate keeps LUT/BRAM/URAM below device limits and removes the baseline DSP pressure. Remaining qualification is Hermes readout, backpressure, and real-trigger behavior, not FPGA fit or route legality.